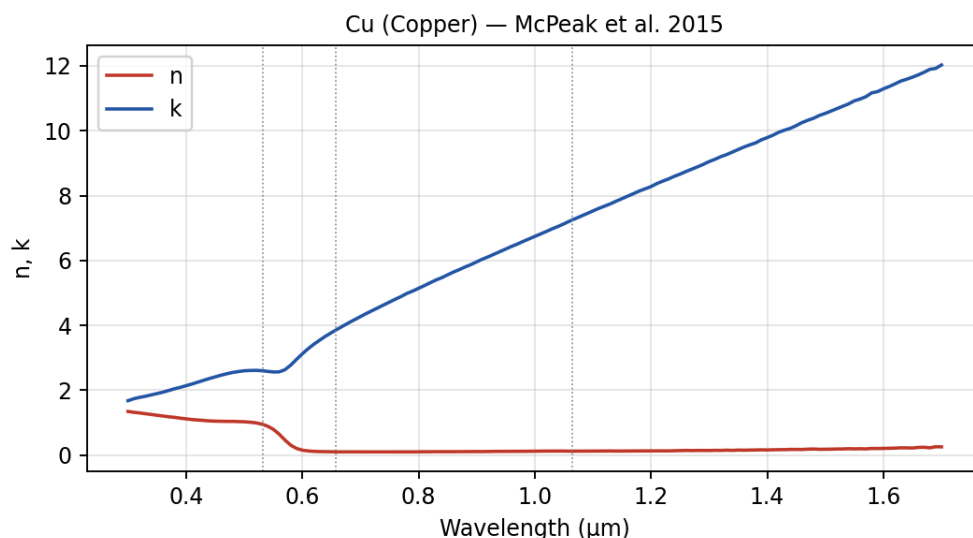


Optical constants of Cu (Copper)

McPeak et al. 2015: n, k 0.3–1.7 μm

Source: <https://refractiveindex.info/?shelf=main&book=Cu&page=McPeak> · Database record: database/data/main/Cu/nk/McPeak.yml (public domain, CC0 1.0)

Complex refractive index (n + ik)



[재현 그림] 데이터베이스 레코드의 표 값으로 다시 그린 n, k 곡선 (원 페이지의 대화형 그래프에 해당). 점선: 532, 658, 1064 nm.

페이지 기본 표시값 ($\lambda = 0.3 \mu\text{m}$): $n = 1.3475$, $k = 1.6794$ · $\epsilon_1 = -1.0048$, $\epsilon_2 = 4.5259$ · $\alpha = 7.0347 \times 10^5 \text{ cm}^{-1}$ · $dn/d\lambda = -2.5987 \mu\text{m}^{-1}$ · group index $n_g = 2.1271$.

Comments

Thermally evaporated Cu at 35 Å/s with base pressure of 3e-8 Torr, template stripped from Si wafer

References

K. M. McPeak, S. V. Jayanti, S. J. P. Kress, S. Meyer, S. Iotti, A. Rossinelli, D. J. Norris. Plasmonic films can easily be better: Rules and recipes. *ACS Photonics* **2**, 326–333 (2015). <https://doi.org/10.1021/ph5004237>

주요 레이저 파장에서의 값 (계산)

λ (nm)	n	k	α (cm ⁻¹)	1/ α (nm)	R (수직 입사)
532	0.9453	2.5997	6.141e+05	16.3	0.6413
658	0.1029	3.8588	7.370e+05	13.6	0.9745
1064	0.1235	7.2450	8.557e+05	11.7	0.9908

원문에 없는 계산값. n, k는 아래 표에서 선형 보간(10 nm 간격 데이터), $\alpha = 4\pi k/\lambda$, $R = [(n-1)^2 + k^2] / [(n+1)^2 + k^2]$ (진공/공기에서 수직 입사, 벌크 가정).

Data (tabulated n, k)

λ (μm)	n	k	λ (μm)	n	k	λ (μm)	n	k
0.3	1.347459987	1.679419071	0.77	0.101672055	4.890330992	1.24	0.135380438	8.594668572
0.31	1.321473211	1.740141215	0.78	0.104166566	4.985764803	1.25	0.140947312	8.661747292
0.32	1.301896917	1.781554261	0.79	0.10154611	5.058785587	1.26	0.145305612	8.740887468
0.33	1.278815346	1.816251273	0.8	0.105089997	5.141307607	1.27	0.142654978	8.812170586
0.34	1.257856058	1.857525737	0.81	0.105640925	5.225721003	1.28	0.145808605	8.885620977
0.35	1.229714372	1.895968733	0.82	0.1047717	5.314412207	1.29	0.143636497	8.96290307
0.36	1.205793784	1.941169403	0.83	0.108065424	5.399044187	1.3	0.14357746	9.049122305
0.37	1.183134074	1.99326522	0.84	0.106329275	5.471682183	1.31	0.148155507	9.118823933
0.38	1.16577487	2.046321345	0.85	0.106803015	5.558363688	1.32	0.146774504	9.199486005
0.39	1.139929606	2.090129064	0.86	0.10806138	5.64355183	1.33	0.153662011	9.262861977
0.4	1.119339006	2.14224644	0.87	0.109423947	5.718126756	1.34	0.14795299	9.342156608
0.41	1.097661459	2.193481406	0.88	0.109226995	5.799390531	1.35	0.156618782	9.420006871
0.42	1.082884327	2.251163803	0.89	0.110739052	5.87326682	1.36	0.153286699	9.497236272
0.43	1.067185209	2.306769228	0.9	0.111474071	5.957887752	1.37	0.158020584	9.563627476
0.44	1.056310845	2.361946782	0.91	0.110565481	6.04212088	1.38	0.159435601	9.624262035
0.45	1.048210496	2.413637347	0.92	0.114354773	6.114742596	1.39	0.164929309	9.719012382
0.46	1.044058354	2.464134299	0.93	0.115741932	6.194384247	1.4	0.160903598	9.787537482
0.47	1.040826414	2.50896784	0.94	0.116243076	6.272201563	1.41	0.165878545	9.863240628
0.48	1.040383818	2.549587906	0.95	0.11651561	6.352367739	1.42	0.170083525	9.953638953
0.49	1.035622719	2.577676166	0.96	0.117227546	6.433641449	1.43	0.170430805	10.01843534
0.5	1.0292166	2.600958825	0.97	0.117612103	6.506983372	1.44	0.177719488	10.07430316
0.51	1.01596237	2.610628188	0.98	0.1200286	6.588429772	1.45	0.176808354	10.15732549
0.52	0.995463808	2.613856957	0.99	0.120693135	6.665250695	1.46	0.174975159	10.25088368
0.53	0.957525814	2.60358516	1	0.123314675	6.744939326	1.47	0.184898936	10.3234775
0.54	0.896412084	2.584135179	1.01	0.123500553	6.819402641	1.48	0.189874398	10.3863397
0.55	0.79745994	2.56420404	1.02	0.12589253	6.894769679	1.49	0.180372997	10.47246554
0.56	0.649913539	2.566649101	1.03	0.126656415	6.97753838	1.5	0.182478591	10.53438648
0.57	0.467667795	2.633707115	1.04	0.125456304	7.048669327	1.51	0.18486728	10.60679857
0.58	0.308052581	2.774526337	1.05	0.127879983	7.128558991	1.52	0.190872505	10.67597454
0.59	0.206477543	2.953105649	1.06	0.123044899	7.21443969	1.53	0.190985335	10.74925464
0.6	0.15342929	3.124794481	1.07	0.124058562	7.290911502	1.54	0.1989454	10.8263373
0.61	0.129738592	3.28082796	1.08	0.12724857	7.368255709	1.55	0.194332409	10.92097821
0.62	0.116677068	3.422223479	1.09	0.126108804	7.447140621	1.56	0.198165888	10.97767142
0.63	0.110069919	3.546563885	1.1	0.127049082	7.525831613	1.57	0.192594699	11.04885029
0.64	0.107194012	3.666809315	1.11	0.127924195	7.605285307	1.58	0.207420155	11.16839394

0.65	0.104232496	3.775693898	1.12	0.12687364	7.67574732	1.59	0.204793341	11.20544244
0.66	0.102539467	3.879628119	1.13	0.129233188	7.744914539	1.6	0.209186703	11.29087061
0.67	0.102449402	3.981770445	1.14	0.126930498	7.827905937	1.61	0.213365941	11.36334128
0.68	0.101216009	4.082308744	1.15	0.128135739	7.904898403	1.62	0.217596043	11.4432228
0.69	0.101603953	4.175083635	1.16	0.131236652	7.984812024	1.63	0.226803672	11.53045706
0.7	0.101236908	4.27062629	1.17	0.130534766	8.061453921	1.64	0.225634749	11.58753257
0.71	0.101557633	4.365353818	1.18	0.132958217	8.142031558	1.65	0.219982349	11.65025288
0.72	0.101132194	4.453675754	1.19	0.134086968	8.211088802	1.66	0.238848652	11.72164872
0.73	0.100848965	4.541494304	1.2	0.132204904	8.276924437	1.67	0.243445023	11.80081631
0.74	0.100919789	4.632837662	1.21	0.131954806	8.369911115	1.68	0.226005004	11.89332074
0.75	0.101173963	4.718605321	1.22	0.134536782	8.444530783	1.69	0.26002749	11.92047039
0.76	0.101837799	4.806908667	1.23	0.132059034	8.511723656	1.7	0.255894846	12.02603352

데이터베이스 레코드 원문 그대로 141행 (브라우저에서 계산한 해시와 대조해 전 행 일치 확인).